Projections and Linear Systems

Last modified: 2026-10-12 00:32:44 (PDT)

Design Matrix

Definition 0.1 (Design matrix (model matrix))**.** In a regression model with n observations and p predictors, the **design matrix** (or **model matrix**) $\mathbf{X}$ is the $n \times p$ matrix whose i-th row is the covariate vector $\tilde{x}_i^\top$ for observation i:

$$\mathbf{X} = \begin{bmatrix} \tilde{x}_1^\top \\ \tilde{x}_2^\top \\ \vdots \\ \tilde{x}_n^\top \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Remark 0.1 (Products with the design matrix). The product $\mathbf{X}\tilde{\beta}$ collects the values $\tilde{x}_i^\top \tilde{\beta}$ for all n observations into a single $n \times 1$ vector:

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} \tilde{x}_1^\top \tilde{\beta} \\ \vdots \\ \tilde{x}_n^\top \tilde{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from Example 2 in Subspaces and Rank as $\mathbf{X}$, with $n = 3$ observations and $p = 2$ columns: an intercept column and one covariate. With $\tilde{\beta} = (1, 2)$,

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$
$$= \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.$$

The matrix $\mathbf{X}^\top\mathbf{X}$ is $p \times p$ and symmetric, since

$$\begin{aligned}(\mathbf{X}^\top\mathbf{X})^\top &= \mathbf{X}^\top (\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top\mathbf{X}\end{aligned}$$

(Theorem 16 in Matrices). When $\mathbf{X}^\top\mathbf{X}$ is invertible, it appears in the OLS estimator $\widehat{\tilde{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\tilde{y}$.

Theorem 0.1 ($\mathbf{X}^\top\mathbf{X}$ is invertible when $\mathbf{X}$ has full column rank)**.** *If $\mathbf{X}$ is an $n \times p$ matrix with* $\text{rank}(\mathbf{X}) = p$ *(Definition 2 in Subspaces and Rank), then the $p \times p$ matrix $\mathbf{X}^\top\mathbf{X}$ is invertible.*

Proof

Proof. Let $\tilde{c}$ be a vector of length p with $\mathbf{X}^\top\mathbf{X}\tilde{c} = \tilde{0}$. Then

$$\begin{aligned} 0 &= \tilde{c}^\top \mathbf{X}^\top\mathbf{X}\tilde{c} && \text{(multiply } \mathbf{X}^\top\mathbf{X}\tilde{c} = \tilde{0} \text{ on the left by } \tilde{c}^\top) \\ &= (\mathbf{X}\tilde{c})^\top(\mathbf{X}\tilde{c}) && \text{(transpose of a product)} \\ &= \|\mathbf{X}\tilde{c}\|^2 && \text{(definition of the Euclidean norm)} \end{aligned}$$

so $\mathbf{X}\tilde{c} = \tilde{0}$. Because $\mathbf{X}\tilde{c} = c_1(\text{column } 1) + \cdots + c_p(\text{column } p)$ and the columns of $\mathbf{X}$ are linearly independent, $\tilde{c} = \tilde{0}$. So the only solution of $\mathbf{X}^\top\mathbf{X}\tilde{c} = \tilde{0}$ is $\tilde{c} = \tilde{0}$, which says that the columns of the square matrix $\mathbf{X}^\top\mathbf{X}$ are linearly independent, and a square matrix with linearly independent columns is invertible (Banerjee and Roy 2014, Corollary 5.7, p. 143). □

Example 0.1 (Inverting $\mathbf{X}^\top\mathbf{X}$)**.** For the rank-2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ from Example 2 in Subspaces and Rank:

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \qquad (\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},$$

and multiplying the two out gives $\mathbf{I}_2$.

Definition 0.2 (Hat matrix)**.** For an $n \times p$ design matrix $\mathbf{X}$ (Definition 0.1) with $\operatorname{rank}(\mathbf{X}) = p$, the **hat matrix** is the $n \times n$ matrix

$$\underset{n\times n}{\mathbf{H}} \overset{\text{def}}{=} \underset{n\times p}{\mathbf{X}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p\times p} \underset{p\times n}{\mathbf{X}^\top}$$

The inverse exists by Theorem 0.1.

Example 0.2 (The hat matrix of an intercept-only model)**.** With $n = 2$ observations and only an intercept, $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (2×1, rank 1), so $\mathbf{X}^\top \mathbf{X} = 2$ and

$$\begin{aligned} \mathbf{H} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}. \end{aligned}$$

Then $\mathbf{H}\tilde{y} = (\bar{y}, \bar{y})$, where $\bar{y} = (y_1 + y_2)/2$: the fitted values of an intercept-only model are the sample mean.

Theorem 0.2 (Hat matrix is a projection matrix)**.** *If $\mathbf{X}$ is an $n \times p$ design matrix with $\operatorname{rank}(\mathbf{X}) = p$, then the hat matrix $\mathbf{H}$ (Definition 0.2) is an orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions).*

Proof

Proof. We verify symmetry and idempotency. Both use that $\mathbf{X}^\top \mathbf{X}$ is symmetric,

$$\begin{aligned} \left(\mathbf{X}^\top \mathbf{X}\right)^\top &= \mathbf{X}^\top \left(\mathbf{X}^\top\right)^\top \\ &= \mathbf{X}^\top \mathbf{X} \end{aligned}$$

(Theorem 16 in Matrices), so its inverse is symmetric too (Corollary 1 in Special Matrices and Decompositions).

Symmetry:

$$\begin{aligned}
\mathbf{H}^\top &= \left(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\right)^\top && \text{(definition of } \mathbf{H}\text{)} \\
&= (\mathbf{X}^\top)^\top \cdot \left((\mathbf{X}^\top\mathbf{X})^{-1}\right)^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\
&= \mathbf{X} \cdot \left((\mathbf{X}^\top\mathbf{X})^{-1}\right)^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\
&= \mathbf{X} \cdot (\mathbf{X}^\top\mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H}\text{)}
\end{aligned}$$

Idempotency:

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(definition of } \mathbf{H}\text{)} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X})(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
&= \mathbf{X}\,\mathbf{I}_p\,(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(definition of the inverse)} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top && \text{(identity matrix)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H}\text{)}
\end{aligned}$$

□

Example 0.3 (The intercept-only hat matrix is a projection)**.** For $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ from Example 0.2, $\mathbf{H}^\top = \mathbf{H}$, and

$$\begin{aligned}
\mathbf{H}^2 &= \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\
&= \mathbf{H},
\end{aligned}$$

as Theorem 0.2 says.

Remark 0.2 (Why it is called the hat matrix). The hat matrix gives the fitted values in linear regression:

$$\begin{aligned}
\hat{\tilde{y}} &= \mathbf{X}\hat{\tilde{\beta}} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\tilde{y} \\
&= \mathbf{H}\tilde{y}.
\end{aligned}$$

Multiplying by $\mathbf{H}$ "puts a hat" on $\tilde{y}$, which is where the name comes from. In Example 0.2, $\mathbf{H}$ puts a hat on $\tilde{y}$ by replacing each y_i with the sample mean $\bar{y}$.

Orthogonal projection onto a subspace

> **Source**
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> This section is adapted from the second half of Zhou (2024c), used under the MIT License (see the license text in Section 2 in Subspaces and Rank). It connects the orthogonal projection matrices of Definition 8 in Special Matrices and Decompositions, defined by an algebraic property, to the geometric decomposition of Theorem 8 in Direct Sums and Orthogonal Complements.

Definition 0.3 (Orthogonal projection onto a subspace)**.** Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ and $\tilde{y} \in \mathbb{R}^p$. By Theorem 8 in Direct Sums and Orthogonal Complements, $\tilde{y} = \tilde{u} + \tilde{v}$ for exactly one $\tilde{u} \in \mathcal{S}$ and $\tilde{v} \in \mathcal{S}^\perp$. The vector $\tilde{u}$ is the **orthogonal projection** of $\tilde{y}$ onto $\mathcal{S}$. It is the projection of $\tilde{y}$ onto $\mathcal{S}$ along $\mathcal{S}^\perp$ (Definition 3 in Direct Sums and Orthogonal Complements).

Example 0.4 (Projecting onto a line in $\mathbb{R}^3$)**.** In Example 15 in Direct Sums and Orthogonal Complements, $\tilde{y} = (3, 1, 2)$ splits as $(2, 2, 0) + (1, -1, 2)$, with $(2, 2, 0) \in \mathcal{S} = \text{span}\{(1, 1, 0)\}$ and $(1, -1, 2) \in \mathcal{S}^\perp$. So the orthogonal projection of $(3, 1, 2)$ onto $\mathcal{S}$ is $(2, 2, 0)$. A vector already in $\mathcal{S}$, such as $(5, 5, 0)$, is its own projection, since $(5, 5, 0) = (5, 5, 0) + \tilde{0}$ and $\tilde{0} \in \mathcal{S}^\perp$ (Theorem 6 in Direct Sums and Orthogonal Complements, Theorem 1 in Subspaces and Rank); a vector of $\mathcal{S}^\perp$, such as $(1, -1, 2)$, projects to $\tilde{0}$, since $(1, -1, 2) = \tilde{0} + (1, -1, 2)$ and $\tilde{0} \in \mathcal{S}$ (Theorem 1 in Subspaces and Rank).

Example 0.5 (A split that is not the orthogonal one)**.** $(3, 1, 2) = (3, 3, 0) + (0, -2, 2)$ also writes $(3, 1, 2)$ as a vector of $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ plus a remainder, but $(1, 1, 0) \cdot (0, -2, 2) = -2 \neq 0$, so the remainder is not in $\mathcal{S}^\perp$, and $(3, 3, 0)$ is not the orthogonal projection of $(3, 1, 2)$ onto $\mathcal{S}$.

Theorem 0.3 (The orthogonal projection is the closest point)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$, let $\tilde{y} \in \mathbb{R}^p$, and let $\tilde{u}$ be the orthogonal projection of $\tilde{y}$ onto $\mathcal{S}$ (Definition 0.3). Then for every $\tilde{w} \in \mathcal{S}$,*

$$\|\tilde{y} - \tilde{u}\| \leq \|\tilde{y} - \tilde{w}\|,$$

with equality only when $\tilde{w} = \tilde{u}$.

Proof

Proof. By Definition 0.3, $\tilde{y} - \tilde{u} \in \mathcal{S}^{\perp}$. Both $\tilde{u}$ and $\tilde{w}$ are in $\mathcal{S}$, so $\tilde{u} - \tilde{w} = \tilde{u} + (-1)\,\tilde{w} \in \mathcal{S}$ (Definition 4 in Subspaces and Rank), and therefore $(\tilde{y} - \tilde{u}) \perp (\tilde{u} - \tilde{w})$ (Definition 4 in Direct Sums and Orthogonal Complements). Then

$$\begin{aligned}
\|\tilde{y} - \tilde{w}\|^2 &= \|(\tilde{y} - \tilde{u}) + (\tilde{u} - \tilde{w})\|^2 && \text{(add and subtract } \tilde{u}\text{)} \\
&= \|\tilde{y} - \tilde{u}\|^2 + \|\tilde{u} - \tilde{w}\|^2 && \text{(Pythagorean theorem, Theorem 2 in Inner Products and Orthogonality)} \\
&\geq \|\tilde{y} - \tilde{u}\|^2, && (\|\tilde{u} - \tilde{w}\|^2 \geq 0\text{, Theorem 1 in Inner Products and Orthogonality})
\end{aligned}$$

with equality exactly when $\|\tilde{u} - \tilde{w}\| = 0$, that is, when $\tilde{w} = \tilde{u}$ (Theorem 1 in Inner Products and Orthogonality, part 1). Taking nonnegative square roots preserves the inequality. □

Example 0.6 (No point of the line is closer)**.** In Example 0.4, $\tilde{y} = (3, 1, 2)$ has projection $\tilde{u} = (2, 2, 0)$ onto $\mathcal{S} = \text{span}\{(1, 1, 0)\}$, at distance $\|(1, -1, 2)\| = \sqrt{6}$. Another point of $\mathcal{S}$, such as $\tilde{w} = (3, 3, 0)$, is farther:

$$\begin{aligned}
\|\tilde{y} - \tilde{w}\| &= \|(0, -2, 2)\| \\
&= \sqrt{8} \\
&> \sqrt{6},
\end{aligned}$$

and the difference of squares is

$$\begin{aligned}
\|\tilde{u} - \tilde{w}\|^2 &= \|(-1, -1, 0)\|^2 \\
&= 2 \\
&= 8 - 6,
\end{aligned}$$

by the Pythagorean theorem (Theorem 2 in Inner Products and Orthogonality), since $\tilde{y} - \tilde{u} \in \mathcal{S}^{\perp}$ and $\tilde{u} - \tilde{w} \in \mathcal{S}$.

Theorem 0.4 (Projecting with an orthonormal basis)**.** *Let $\mathcal{S}$ be a subspace of $\mathbb{R}^p$ with $\dim(\mathcal{S}) = r \geq 1$, and let $\mathbf{Q}$ be the $p \times r$ matrix whose columns $\tilde{q}_1, \ldots, \tilde{q}_r$ are an orthonormal basis of $\mathcal{S}$ (Definition 9 in Inner Products and Orthogonality, Corollary 1 in Inner Products and Orthogonality). Then for every $\tilde{y} \in \mathbb{R}^p$, $\mathbf{Q}\mathbf{Q}^{\top}\tilde{y}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{S}$, and $\mathbf{Q}\mathbf{Q}^{\top}$ is an orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions).*

Proof

Proof. $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r$**.** Write q_{ki} for entry k of $\tilde{q}_i$, which is entry (k, i) of $\mathbf{Q}$. Then

$$\begin{aligned}(\mathbf{Q}^\top\mathbf{Q})_{ij} &= \sum_{k=1}^{p}(\mathbf{Q}^\top)_{ik}\, q_{kj} && \text{(Definition 7 in Matrices)}\\ &= \sum_{k=1}^{p} q_{ki}\, q_{kj} && \text{(Definition 3 in Matrices)}\\ &= \tilde{q}_i \cdot \tilde{q}_j, && \text{(Definition 7 in Vectors)}\end{aligned}$$

which is 1 if $i = j$ and 0 otherwise (Definition 17 in Vectors), so $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r$ (Definition 10 in Matrices).

The split. Let $\tilde{u} = \mathbf{Q}\mathbf{Q}^\top\tilde{y}$ and $\tilde{v} = \tilde{y} - \tilde{u}$, so $\tilde{y} = \tilde{u} + \tilde{v}$. $\tilde{u} = \mathbf{Q}\,(\mathbf{Q}^\top\tilde{y})$ (Theorem 5 in Matrices) is in $\mathcal{C}(\mathbf{Q})$ (Definition 1 in Column Space, Null Space and Rank-Nullity), which is $\text{span}\{\tilde{q}_1, \ldots, \tilde{q}_r\} = \mathcal{S}$ (Theorem 1 in Column Space, Null Space and Rank-Nullity). For $\tilde{v}$,

$$\begin{aligned}\mathbf{Q}^\top\tilde{v} &= \mathbf{Q}^\top(\tilde{y} - \mathbf{Q}\mathbf{Q}^\top\tilde{y}) && \text{(substitute)}\\ &= \mathbf{Q}^\top\tilde{y} - \mathbf{Q}^\top(\mathbf{Q}\mathbf{Q}^\top\tilde{y}) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1)\\ &= \mathbf{Q}^\top\tilde{y} - (\mathbf{Q}^\top\mathbf{Q})\,\mathbf{Q}^\top\tilde{y} && \text{(Theorem 5 in Matrices)}\\ &= \mathbf{Q}^\top\tilde{y} - \mathbf{I}_r\,\mathbf{Q}^\top\tilde{y} && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r\text{, from the first part of the proof})\\ &= \mathbf{Q}^\top\tilde{y} - \mathbf{Q}^\top\tilde{y} && \text{(Theorem 7 in Matrices)}\\ &= \tilde{0}_r, && \text{(arithmetic)}\end{aligned}$$

so $\tilde{v} \in \mathcal{N}(\mathbf{Q}^\top)$, and

$$\begin{aligned}\mathcal{N}(\mathbf{Q}^\top) &= \mathcal{C}(\mathbf{Q})^\perp\\ &= \mathcal{S}^\perp\end{aligned}$$

(Theorem 7 in Direct Sums and Orthogonal Complements). By the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements, $\tilde{u}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{S}$.

$\mathbf{Q}\mathbf{Q}^\top$ is symmetric and idempotent.

$$\begin{aligned}(\mathbf{Q}\mathbf{Q}^\top)^\top &= (\mathbf{Q}^\top)^\top\,\mathbf{Q}^\top && \text{(Theorem 16 in Matrices)}\\ &= \mathbf{Q}\mathbf{Q}^\top, && \text{(transposing twice, Definition 3 in Matrices)}\end{aligned}$$

and

$$\begin{aligned}(\mathbf{Q}\mathbf{Q}^\top)(\mathbf{Q}\mathbf{Q}^\top) &= \mathbf{Q}\,(\mathbf{Q}^\top\mathbf{Q})\,\mathbf{Q}^\top && \text{(Theorem 5 in Matrices)}\\ &= \mathbf{Q}\,\mathbf{I}_r\,\mathbf{Q}^\top && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_r\text{, from the first part of the proof})\\ &= \mathbf{Q}\mathbf{Q}^\top. && \text{(Theorem 7 in Matrices)}\end{aligned}$$

□

Example 0.7 (The projection matrix onto a line)**.** For $\mathcal{S} = \text{span}\{(1,1,0)\}$, the single vector $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$ is an orthonormal basis: it spans $\mathcal{S}$, it is nonzero and so linearly independent, and

$$\begin{aligned}\|\tilde{q}_1\| &= \sqrt{\tfrac{1}{2} + \tfrac{1}{2} + 0} \\ &= 1.\end{aligned}$$

With $\mathbf{Q} = [\tilde{q}_1]$,

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \tfrac{1}{2}\begin{bmatrix}1\\1\\0\end{bmatrix}\begin{bmatrix}1 & 1 & 0\end{bmatrix} \\ &= \begin{bmatrix}\frac{1}{2} & \frac{1}{2} & 0\\ \frac{1}{2} & \frac{1}{2} & 0\\ 0 & 0 & 0\end{bmatrix}.\end{aligned}$$

Applied to $\tilde{y} = (3,1,2)$, it gives $\left(\frac{3+1}{2}, \frac{3+1}{2}, 0\right) = (2,2,0)$, the projection found in Example 0.4.

Theorem 0.5 (A projection matrix projects onto its column space, and is the only one that does)**.**

1. *If* $\mathbf{P}$ *is a* $p \times p$ *orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions), then for every* $\tilde{y} \in \mathbb{R}^p$, $\mathbf{P}\tilde{y}$ *is the orthogonal projection of* $\tilde{y}$ *onto* $\mathcal{C}(\mathbf{P})$ *(Definition 0.3).*
2. *If* $\mathbf{P}_1$ *and* $\mathbf{P}_2$ *are* $p \times p$ *matrices such that, for every* $\tilde{y}$*, both* $\mathbf{P}_1\tilde{y}$ *and* $\mathbf{P}_2\tilde{y}$ *are the orthogonal projection of* $\tilde{y}$ *onto the same subspace* $\mathcal{S}$*, then* $\mathbf{P}_1 = \mathbf{P}_2$*.*

Proof

Proof. **Part 1.** Write $\tilde{y} = \mathbf{P}\tilde{y} + (\tilde{y} - \mathbf{P}\tilde{y})$. The first term is in $\mathcal{C}(\mathbf{P})$ (Definition 1 in Column Space, Null Space and Rank-Nullity). For the second,

$$\begin{aligned}
\mathbf{P}\left(\tilde{y} - \mathbf{P}\tilde{y}\right) &= \mathbf{P}\tilde{y} - \mathbf{P}\left(\mathbf{P}\tilde{y}\right) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1\text{)} \\
&= \mathbf{P}\tilde{y} - \left(\mathbf{PP}\right)\tilde{y} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{P}\tilde{y} - \mathbf{P}^2\tilde{y} && \text{(Definition 2 in Special Matrices and Decompositions)} \\
&= \mathbf{P}\tilde{y} - \mathbf{P}\tilde{y} && \text{(}\mathbf{P}\text{ is idempotent)} \\
&= \tilde{0}_p, && \text{(arithmetic)}
\end{aligned}$$

so $\tilde{y} - \mathbf{P}\tilde{y} \in \mathcal{N}(\mathbf{P})$. Since $\mathbf{P} = \mathbf{P}^\top$,

$$\begin{aligned}
\mathcal{N}(\mathbf{P}) &= \mathcal{N}(\mathbf{P}^\top) \\
&= \mathcal{C}(\mathbf{P})^\perp
\end{aligned}$$

(Theorem 7 in Direct Sums and Orthogonal Complements). By the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements, $\mathbf{P}\tilde{y}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{C}(\mathbf{P})$.

Part 2. The orthogonal projection of a vector onto $\mathcal{S}$ is unique (Definition 0.3), so $\mathbf{P}_1\tilde{e}_j = \mathbf{P}_2\tilde{e}_j$ for each indicator vector $\tilde{e}_j$ (Definition 12 in Vectors). $\mathbf{P}_1\tilde{e}_j$ is column j of $\mathbf{P}_1$: by Theorem 3 in Subspaces and Rank it is the combination of the columns of $\mathbf{P}_1$ with coefficients the entries of $\tilde{e}_j$, which are 1 in place j and 0 elsewhere (Definition 12 in Vectors). Likewise for $\mathbf{P}_2$, so the two matrices have the same columns. □

Example 0.8 (Two routes to the same projection matrix)**.** The matrix $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ of Example 10 in Special Matrices and Decompositions is an orthogonal projection matrix, and its column space is span$\{(1,0)\}$, the horizontal axis. By part 1, $\mathbf{P}(v_1, v_2) = (v_1, 0)$ is the orthogonal projection onto that axis: the remainder $(0, v_2)$ is orthogonal to $(1,0)$. The vector $\tilde{q}_1 = (1,0)$ has length 1 and spans the axis, so it is an orthonormal basis of it, and with $\mathbf{B} = [\tilde{q}_1]$,

$$\begin{aligned}
\mathbf{B}\mathbf{B}^\top &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\
&= \mathbf{P},
\end{aligned}$$

as part 2 and Theorem 0.4 require.

The oblique projection $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of Example 10 in Special Matrices and Decompositions also maps every vector into the horizontal axis. It sends $(0,1)$ to $(1,0)$, but the remainder $(0,1) - (1,0) = (-1,1)$ is not orthogonal to the axis, so $(1,0)$ is not the orthogonal projection of $(0,1)$ onto the axis, which is $\mathbf{P}(0,1) = (0,0)$.

Theorem 0.6 (The hat matrix projects onto the column space of the design matrix)**.** *If* $\mathbf{X}$ *is an* $n \times p$ *design matrix with* $\text{rank}(\mathbf{X}) = p$*, then for every* $\tilde{y} \in \mathbb{R}^n$*,* $\mathbf{H}\tilde{y}$ *(Definition 0.2) is the orthogonal projection of* $\tilde{y}$ *onto* $\mathcal{C}(\mathbf{X})$*. So* $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$ *for any* $n \times p$ *matrix* $\mathbf{Q}$ *whose columns are an orthonormal basis of* $\mathcal{C}(\mathbf{X})$*.*

Proof

Proof. $\mathbf{H}$ is an orthogonal projection matrix (Theorem 0.2), so $\mathbf{H}\tilde{y}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{C}(\mathbf{H})$ (Theorem 0.5, part 1). It remains to show $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$.
$\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$**.** $\mathbf{H}\tilde{z} = \mathbf{X}\left((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\tilde{z}\right)$ (Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{X})$ (Definition 1 in Column Space, Null Space and Rank-Nullity).
$\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$**.** First,

$$\begin{aligned}
\mathbf{H}\mathbf{X} &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X} && \text{(Definition 0.2)}\\
&= \mathbf{X}\left((\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X})\right) && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{X}\,\mathbf{I}_p && \text{(Definition 5 in Special Matrices and Decompositions)}\\
&= \mathbf{X}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

So for every $\tilde{b} \in \mathbb{R}^p$, $\mathbf{X}\tilde{b} = (\mathbf{H}\mathbf{X})\,\tilde{b}$, since $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $(\mathbf{H}\mathbf{X})\,\tilde{b} = \mathbf{H}\,(\mathbf{X}\tilde{b})$ (Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{H})$.
$\mathcal{C}(\mathbf{X})$ has dimension $p \geq 1$ (Theorem 5 in Column Space, Null Space and Rank-Nullity), so an orthonormal basis of it has p vectors (Definition 8 in Subspaces and Rank), and $\mathbf{Q}\mathbf{Q}^\top$ also gives the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ (Theorem 0.4, with n in place of p and $r = p$). By Theorem 0.5, part 2, $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$. □

Example 0.9 (The intercept-only hat matrix, from an orthonormal basis)**.** For the intercept-only design $\mathbf{X} = \begin{bmatrix}1\\1\end{bmatrix}$ of Example 0.2, $\mathcal{C}(\mathbf{X}) = \text{span}\{(1,1)\}$, with orthonormal basis $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1,1)$, since

$$\begin{aligned}
\|\tilde{q}_1\| &= \sqrt{\tfrac{1}{2} + \tfrac{1}{2}}\\
&= 1.
\end{aligned}$$

With $\mathbf{Q} = [\tilde{q}_1]$,

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \tfrac{1}{2}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix}\\
&= \begin{bmatrix}0.5 & 0.5\\0.5 & 0.5\end{bmatrix},
\end{aligned}$$

the hat matrix found in Example 0.2. The fitted values $(\bar{y}, \bar{y})$ are the closest point to $\tilde{y}$ on the line of constant vectors (Theorem 0.3), and the residuals $(y_1 - \bar{y}, y_2 - \bar{y})$ are orthogonal to $(1, 1)$:

$$\begin{aligned}(1,1) \cdot (y_1 - \bar{y}, y_2 - \bar{y}) &= y_1 + y_2 - 2\bar{y} \\ &= 0,\end{aligned}$$

since $2\bar{y} = y_1 + y_2$.

Generalized inverses and linear systems

Source

This section is adapted from Zhou (2024b), used under the MIT License (see the license text in Section 2 in Subspaces and Rank). The source defers the existence of the Moore-Penrose inverse to the singular value decomposition; the proof here builds it from a rank factorization (Theorem 12 in Column Space, Null Space and Rank-Nullity) instead, and the section adds the projection $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ for any generalized inverse $\mathbf{G}$ of $\mathbf{X}^\top\mathbf{X}$, which the source does not cover.

Theorem 0.7 (A scalar factor passes through a matrix product)**.** *For an $m \times k$ matrix $\mathbf{A}$, a $k \times n$ matrix $\mathbf{B}$ and a number c,*

$$\begin{aligned}\mathbf{A}\,(c\,\mathbf{B}) &= c\,(\mathbf{A}\mathbf{B}) \\ &= (c\,\mathbf{A})\,\mathbf{B}.\end{aligned}$$

In particular, $\mathbf{A}\,(\mathbf{B} - \mathbf{C}) = \mathbf{A}\mathbf{B} - \mathbf{A}\mathbf{C}$ and $(\mathbf{B} - \mathbf{C})\,\mathbf{D} = \mathbf{B}\mathbf{D} - \mathbf{C}\mathbf{D}$ for matrices of compatible sizes, where $\mathbf{B} - \mathbf{C} = \mathbf{B} + (-1)\,\mathbf{C}$ (Theorem 4 in Matrices, Definition 6 in Matrices).

Proof

Proof. **$\mathbf{A}\,(c\,\mathbf{B}) = c\,(\mathbf{A}\mathbf{B})$.** For each entry (i, j),

$$
\begin{aligned}
(\mathbf{A}\,(c\,\mathbf{B}))_{ij} &= \sum_{l=1}^{k} a_{il}\,(c\,\mathbf{B})_{lj} && \text{(Definition 7 in Matrices)}\\
&= \sum_{l=1}^{k} a_{il}\,(c\,b_{lj}) && \text{(Definition 6 in Matrices)}\\
&= \sum_{l=1}^{k} c\,(a_{il}\,b_{lj}) && \text{(reorder the product of numbers)}\\
&= c\sum_{l=1}^{k} a_{il}\,b_{lj} && \text{(factor } c \text{ out of the sum)}\\
&= c\,(\mathbf{AB})_{ij} && \text{(Definition 7 in Matrices)}\\
&= (c\,(\mathbf{AB}))_{ij}. && \text{(Definition 6 in Matrices)}
\end{aligned}
$$

$(c\,\mathbf{A})\,\mathbf{B} = c\,(\mathbf{AB})$**.** For each entry (i, j),

$$
\begin{aligned}
((c\,\mathbf{A})\,\mathbf{B})_{ij} &= \sum_{l=1}^{k} (c\,a_{il})\,b_{lj} && \text{(Definition 7 in Matrices, Definition 6 in Matrices)}\\
&= c\sum_{l=1}^{k} a_{il}\,b_{lj} && \text{(factor } c \text{ out of the sum)}\\
&= (c\,(\mathbf{AB}))_{ij}. && \text{(Definition 7 in Matrices, Definition 6 in Matrices)}
\end{aligned}
$$

Differences.

$$
\begin{aligned}
\mathbf{A}\,(\mathbf{B}-\mathbf{C}) &= \mathbf{A}\,(\mathbf{B}+(-1)\,\mathbf{C}) && \text{(the difference as a sum)}\\
&= \mathbf{AB}+\mathbf{A}\,((-1)\,\mathbf{C}) && \text{(Theorem 6 in Matrices)}\\
&= \mathbf{AB}+(-1)\,(\mathbf{AC}) && \text{(first part)}\\
&= \mathbf{AB}-\mathbf{AC}, && \text{(the difference as a sum)}
\end{aligned}
$$

and $(\mathbf{B}-\mathbf{C})\,\mathbf{D} = \mathbf{BD}-\mathbf{CD}$ the same way, using the second law in Theorem 6 in Matrices and the second part. A column vector is a matrix with one column, so all of this applies with $\mathbf{B}$, $\mathbf{C}$ or $\mathbf{D}$ a vector. □

Example 0.10 (Pulling a scalar out of a product)**.** With $\mathbf{A} = \begin{bmatrix}1 & 2\end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix}3\\4\end{bmatrix}$:

$$
\begin{aligned}
\mathbf{A}\,(2\,\mathbf{B}) &= \begin{bmatrix}1 & 2\end{bmatrix}\begin{bmatrix}6\\8\end{bmatrix}\\
&= 22,
\end{aligned}
$$

and

$$\begin{aligned} 2\,(\mathbf{AB}) &= 2\,(3+8) \\ &= 22. \end{aligned}$$

Definition 0.4 (Generalized inverse)**.** A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ such that

$$\underset{m\times n}{\mathbf{A}}\ \underset{n\times m}{\mathbf{G}}\ \underset{m\times n}{\mathbf{A}} = \mathbf{A}.$$

Example 0.11 (A matrix with many generalized inverses)**.** Let $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{G} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$. Then

$$\begin{aligned} \mathbf{AGA} &= \begin{bmatrix} g_{11} & g_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} && \text{(Definition 7 in Matrices, for } \mathbf{AG}) \\ &= \begin{bmatrix} g_{11} & 0 \\ 0 & 0 \end{bmatrix}, && \text{(Definition 7 in Matrices)} \end{aligned}$$

which equals $\mathbf{A}$ exactly when $g_{11} = 1$. So every matrix $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ is a generalized inverse of $\mathbf{A}$: a generalized inverse need not be unique. The zero matrix is not one, since its $(1,1)$ entry is 0.

Theorem 0.8 (An invertible matrix has only one generalized inverse)**.** *If $\mathbf{A}$ is an invertible $p \times p$ matrix (Definition 6 in Special Matrices and Decompositions), its only generalized inverse is $\mathbf{A}^{-1}$.*

Proof

Proof.

$$\begin{aligned} \mathbf{A}\mathbf{A}^{-1}\mathbf{A} &= \mathbf{I}_p\mathbf{A} \\ &= \mathbf{A} \end{aligned}$$

(Definition 5 in Special Matrices and Decompositions, Theorem 7 in Matrices), so $\mathbf{A}^{-1}$ is a generalized inverse. If $\mathbf{G}$ is any generalized inverse, then

$$
\begin{aligned}
\mathbf{G} &= \mathbf{I}_p\,\mathbf{G}\,\mathbf{I}_p && \text{(Theorem 7 in Matrices)}\\
&= (\mathbf{A}^{-1}\mathbf{A})\,\mathbf{G}\,(\mathbf{A}\mathbf{A}^{-1}) && \text{(Definition 5 in Special Matrices and Decompositions)}\\
&= \mathbf{A}^{-1}\,(\mathbf{A}\mathbf{G}\mathbf{A})\,\mathbf{A}^{-1} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A}^{-1}\mathbf{A}\mathbf{A}^{-1} && \text{(Definition 0.4)}\\
&= \mathbf{I}_p\,\mathbf{A}^{-1} && \text{(Definition 5 in Special Matrices and Decompositions)}\\
&= \mathbf{A}^{-1}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

□

Example 0.12 (The generalized inverse of an invertible matrix)**.** For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ of Example 5 in Special Matrices and Decompositions, the only generalized inverse is $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$. The matrix of Example 0.11, by contrast, has infinitely many, so by Theorem 0.8 it is singular.

Theorem 0.9 (A generalized inverse solves every solvable system)**.** *Let* $\mathbf{A}$ *be* $m \times n$*, let* $\mathbf{G}$ *be a generalized inverse of* $\mathbf{A}$ *(Definition 0.4), and let* $\tilde{b} \in \mathbb{R}^m$*. If* $\mathbf{A}\tilde{x} = \tilde{b}$ *has a solution, then* $\mathbf{G}\tilde{b}$ *is a solution.*

Proof

Proof. Let $\tilde{x}_0$ be a solution, so $\mathbf{A}\tilde{x}_0 = \tilde{b}$. Then

$$
\begin{aligned}
\mathbf{A}\,(\mathbf{G}\tilde{b}) &= \mathbf{A}\mathbf{G}\,(\mathbf{A}\tilde{x}_0) && \text{(substitute } \tilde{b} = \mathbf{A}\tilde{x}_0)\\
&= (\mathbf{A}\mathbf{G}\mathbf{A})\,\tilde{x}_0 && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A}\tilde{x}_0 && \text{(Definition 0.4)}\\
&= \tilde{b}. && (\tilde{x}_0 \text{ is a solution})
\end{aligned}
$$

□

Example 0.13 (Solving with a generalized inverse)**.** With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and its generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ (Example 0.11), the system $\mathbf{A}\tilde{x} = (2, 0)$ has the solution $(2, 0)$, and $\mathbf{G}\,(2, 0) = (2, 14)$ is a solution too: $\mathbf{A}\,(2, 14) = (2, 0)$. The system $\mathbf{A}\tilde{x} = (2, 1)$ has no solution, since the second entry of $\mathbf{A}\tilde{x}$ is always 0; there $\mathbf{G}\,(2, 1) = (7, 14)$ gives

$\mathbf{A}\,(7,14) = (7,0) \neq (2,1)$.

Definition 0.5 (Moore-Penrose inverse)**.** A **Moore-Penrose inverse** of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ satisfying all four conditions

1. $\mathbf{AGA} = \mathbf{A}$;
2. $\mathbf{GAG} = \mathbf{G}$;
3. $(\mathbf{AG})^\top = \mathbf{AG}$;
4. $(\mathbf{GA})^\top = \mathbf{GA}$.

Example 0.14 (Checking the four conditions)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, the matrix $\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ satisfies all four:

$$\begin{aligned} \mathbf{AG} &= \mathbf{GA} \\ &= \mathbf{A} \end{aligned}$$

and $\mathbf{A}^2 = \mathbf{A}$, so conditions 1 and 2 read $\mathbf{A} = \mathbf{A}$, and conditions 3 and 4 hold because $\mathbf{A}$ is symmetric.

The generalized inverse $\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ of $\mathbf{A}$ (Example 0.11) is not a Moore-Penrose inverse: $\mathbf{AG}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, which is not symmetric, so condition 3 fails.

Theorem 0.10 (A Moore-Penrose inverse is a generalized inverse)**.** *If* $\mathbf{G}$ *is a Moore-Penrose inverse of an* $m \times n$ *matrix* $\mathbf{A}$ *(Definition 0.5), then* $\mathbf{G}$ *is a generalized inverse of* $\mathbf{A}$ *(Definition 0.4).*

Proof

Proof. A generalized inverse of $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{G}$ with $\mathbf{AGA} = \mathbf{A}$. A Moore-Penrose inverse is an $n \times m$ matrix, and condition 1 of Definition 0.5 is $\mathbf{AGA} = \mathbf{A}$. □

Example 0.15 (The Moore-Penrose inverse of a rank-one matrix)**.** The 3×2 matrix $\mathbf{A}$ below has rank 1, because its second column is twice its first, so it has no ordinary inverse. The function `MASS::ginv()` computes its Moore-Penrose inverse $\mathbf{G}$. The code checks the four conditions of Definition 0.5, of which the first is the equation that makes $\mathbf{G}$ a generalized inverse.

```r
A <- cbind(c(1, 2, 3), c(2, 4, 6))
G <- MASS::ginv(A)
conditions <- c(
  "1: AGA = A" = isTRUE(all.equal(A %*% G %*% A, A)),
  "2: GAG = G" = isTRUE(all.equal(G %*% A %*% G, G)),
  "3: AG symmetric" = isTRUE(all.equal(A %*% G, t(A %*% G))),
  "4: GA symmetric" = isTRUE(all.equal(G %*% A, t(G %*% A)))
)
conditions
#>      1: AGA = A      2: GAG = G 3: AG symmetric 4: GA symmetric
#>             TRUE            TRUE            TRUE            TRUE
round(G, 4)
#>        [,1]   [,2]   [,3]
#> [1,] 0.0143 0.0286 0.0429
#> [2,] 0.0286 0.0571 0.0857
```

All 4 of 4 conditions hold, so $\mathbf{G}$ is a Moore-Penrose inverse of $\mathbf{A}$. By Theorem 0.10 it is also a generalized inverse, which is condition 1: $\mathbf{AGA} = \mathbf{A}$ up to rounding error.

Theorem 0.11 (A matrix has at most one Moore-Penrose inverse)**.** *If $\mathbf{G}_1$ and $\mathbf{G}_2$ are both Moore-Penrose inverses of an $m \times n$ matrix $\mathbf{A}$ (Definition 0.5), then $\mathbf{G}_1 = \mathbf{G}_2$.*

Proof

Proof. $\mathbf{AG}_1 = \mathbf{AG}_2$**.**

$$
\begin{aligned}
\mathbf{AG}_1 &= (\mathbf{AG}_1)^\top && \text{(condition 3 for } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_1{}^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{G}_1{}^\top (\mathbf{AG}_2\mathbf{A})^\top && \text{(condition 1 for } \mathbf{G}_2\text{)} \\
&= \mathbf{G}_1{}^\top \mathbf{A}^\top \mathbf{G}_2{}^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{G}_1{}^\top \mathbf{A}^\top)(\mathbf{G}_2{}^\top \mathbf{A}^\top) && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{AG}_1)^\top (\mathbf{AG}_2)^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{AG}_1)(\mathbf{AG}_2) && \text{(condition 3 for } \mathbf{G}_1 \text{ and } \mathbf{G}_2\text{)} \\
&= (\mathbf{AG}_1\mathbf{A})\,\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{AG}_2. && \text{(condition 1 for } \mathbf{G}_1\text{)}
\end{aligned}
$$

$\mathbf{G}_1\mathbf{A} = \mathbf{G}_2\mathbf{A}$**.**

$$
\begin{aligned}
\mathbf{G}_1\mathbf{A} &= (\mathbf{G}_1\mathbf{A})^\top && \text{(condition 4 for } \mathbf{G}_1\text{)} \\
&= \mathbf{A}^\top \mathbf{G}_1{}^\top && \text{(Theorem 16 in Matrices)} \\
&= (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top \mathbf{G}_1{}^\top && \text{(condition 1 for } \mathbf{G}_2\text{)} \\
&= \mathbf{A}^\top \mathbf{G}_2{}^\top \mathbf{A}^\top \mathbf{G}_1{}^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{A}^\top \mathbf{G}_2{}^\top)(\mathbf{A}^\top \mathbf{G}_1{}^\top) && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{G}_2\mathbf{A})^\top (\mathbf{G}_1\mathbf{A})^\top && \text{(Theorem 16 in Matrices, twice)} \\
&= (\mathbf{G}_2\mathbf{A})(\mathbf{G}_1\mathbf{A}) && \text{(condition 4 for } \mathbf{G}_2 \text{ and } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_2(\mathbf{A}\mathbf{G}_1\mathbf{A}) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_2\mathbf{A}. && \text{(condition 1 for } \mathbf{G}_1\text{)}
\end{aligned}
$$

Conclusion.

$$
\begin{aligned}
\mathbf{G}_1 &= \mathbf{G}_1\mathbf{A}\mathbf{G}_1 && \text{(condition 2 for } \mathbf{G}_1\text{)} \\
&= \mathbf{G}_1(\mathbf{A}\mathbf{G}_1) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_1(\mathbf{A}\mathbf{G}_2) && \text{(the first step)} \\
&= (\mathbf{G}_1\mathbf{A})\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{G}_2\mathbf{A})\mathbf{G}_2 && \text{(the second step)} \\
&= \mathbf{G}_2\mathbf{A}\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_2. && \text{(condition 2 for } \mathbf{G}_2\text{)}
\end{aligned}
$$

□

Example 0.16 (Only one of the many generalized inverses qualifies)**.** The matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ has infinitely many generalized inverses $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$ (Example 0.11), and $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ is a Moore-Penrose inverse (Example 0.14). By Theorem 0.11 it is the only one, so every other choice of g_{12}, g_{21}, g_{22} breaks condition 2, 3 or 4. For instance, $g_{21} = 1$ (the rest 0) gives $\mathbf{G}\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, which is not symmetric.

Theorem 0.12 (Every matrix has a Moore-Penrose inverse)**.** *Every $m \times n$ matrix $\mathbf{A}$ has a Moore-Penrose inverse (Definition 0.5), written $\mathbf{A}^+$; it is unique by Theorem 0.11.*

- *If $\mathbf{A} = \mathbf{0}_{m\times n}$, then $\mathbf{A}^+ = \mathbf{0}_{n\times m}$.*
- *If $\operatorname{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A} = \mathbf{C}\mathbf{R}$ is a rank factorization (Definition 7 in Column Space, Null Space and Rank-Nullity), then*

$$\underbrace{\mathbf{A}^{+}}_{n\times m} = \underbrace{\mathbf{R}^{\top}}_{n\times r}\underbrace{(\mathbf{R}\mathbf{R}^{\top})^{-1}}_{r\times r}\underbrace{(\mathbf{C}^{\top}\mathbf{C})^{-1}}_{r\times r}\underbrace{\mathbf{C}^{\top}}_{r\times m}.$$

Proof

Proof. **The two cases cover every matrix.** If $\mathbf{A} \neq \mathbf{0}_{m\times n}$, it has a nonzero column, which is linearly independent on its own ($c\,\tilde{v} = \tilde{0}$ with $\tilde{v} \neq \tilde{0}$ forces $c = 0$), so $\operatorname{rank}(\mathbf{A}) \geq 1$ (Definition 2 in Subspaces and Rank).
The zero matrix. With $\mathbf{A} = \mathbf{0}_{m\times n}$ and $\mathbf{G} = \mathbf{0}_{n\times m}$, every product in conditions 1 to 4 is a zero matrix, and a zero matrix equals its own transpose, so all four conditions hold.
The two inverses exist. Now suppose $\operatorname{rank}(\mathbf{A}) = r \geq 1$. A rank factorization exists (Theorem 12 in Column Space, Null Space and Rank-Nullity). $\mathbf{C}$ is $m \times r$ and $\mathbf{R}$ is $r \times n$. Both have rank r:

$$\begin{aligned}
r &= \operatorname{rank}(\mathbf{C}\mathbf{R}) && (\mathbf{A} = \mathbf{C}\mathbf{R})\\
&\leq \operatorname{rank}(\mathbf{C}) && \text{(Theorem 7 in Column Space, Null Space and Rank-Nullity)}\\
&\leq r, && \text{(Corollary 1 in Column Space, Null Space and Rank-Nullity, } \mathbf{C} \text{ has } r \text{ columns)}
\end{aligned}$$

and, using $(\mathbf{C}\mathbf{R})^{\top} = \mathbf{R}^{\top}\mathbf{C}^{\top}$ (Theorem 16 in Matrices),

$$\begin{aligned}
r &= \operatorname{rank}(\mathbf{R}^{\top}\mathbf{C}^{\top}) && \text{(Theorem 8 in Column Space, Null Space and Rank-Nullity)}\\
&\leq \operatorname{rank}(\mathbf{R}^{\top}) && \text{(Theorem 7 in Column Space, Null Space and Rank-Nullity)}\\
&\leq r. && \text{(Corollary 1 in Column Space, Null Space and Rank-Nullity, } \mathbf{R}^{\top} \text{ has } r \text{ columns)}
\end{aligned}$$

So $\mathbf{C}$ and $\mathbf{R}^{\top}$ have full column rank r, and $\mathbf{C}^{\top}\mathbf{C}$ and $\mathbf{R}\mathbf{R}^{\top} = (\mathbf{R}^{\top})^{\top}\mathbf{R}^{\top}$ are invertible (Theorem 0.1). They are symmetric:

$$\begin{aligned}
(\mathbf{C}^{\top}\mathbf{C})^{\top} &= \mathbf{C}^{\top}(\mathbf{C}^{\top})^{\top}\\
&= \mathbf{C}^{\top}\mathbf{C}
\end{aligned}$$

(Theorem 16 in Matrices, Definition 3 in Matrices), and likewise for $\mathbf{R}\mathbf{R}^{\top}$. Write $\mathbf{S} \overset{\text{def}}{=} (\mathbf{R}\mathbf{R}^{\top})^{-1}$ and $\mathbf{T} \overset{\text{def}}{=} (\mathbf{C}^{\top}\mathbf{C})^{-1}$, both symmetric (Corollary 1 in Special Matrices and Decompositions), and $\mathbf{G} \overset{\text{def}}{=} \mathbf{R}^{\top}\mathbf{S}\mathbf{T}\mathbf{C}^{\top}$.
Two products.

$$\begin{aligned}
\mathbf{A}\mathbf{G} &= \mathbf{C}\mathbf{R}\,\mathbf{R}^{\top}\mathbf{S}\mathbf{T}\mathbf{C}^{\top} && \text{(substitute } \mathbf{A} \text{ and } \mathbf{G})\\
&= \mathbf{C}\,(\mathbf{R}\mathbf{R}^{\top})\,\mathbf{S}\,\mathbf{T}\mathbf{C}^{\top} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{C}\,\mathbf{I}_r\,\mathbf{T}\mathbf{C}^{\top} && \text{(Definition 5 in Special Matrices and Decompositions)}\\
&= \mathbf{C}\mathbf{T}\mathbf{C}^{\top}, && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

$$
\begin{aligned}
\mathbf{GA} &= \mathbf{R}^\top \mathbf{S}\mathbf{T}\mathbf{C}^\top \mathbf{C}\mathbf{R} && \text{(substitute } \mathbf{G} \text{ and } \mathbf{A}\text{)} \\
&= \mathbf{R}^\top \mathbf{S}\,\mathbf{T}\,(\mathbf{C}^\top \mathbf{C})\,\mathbf{R} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}\,\mathbf{I}_r\,\mathbf{R} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{R}^\top \mathbf{S}\mathbf{R}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

Conditions 3 and 4.

$$
\begin{aligned}
(\mathbf{AG})^\top &= (\mathbf{C}\mathbf{T}\mathbf{C}^\top)^\top && \text{(first product)} \\
&= (\mathbf{C}\,(\mathbf{T}\mathbf{C}^\top))^\top && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{T}\mathbf{C}^\top)^\top \mathbf{C}^\top && \text{(Theorem 16 in Matrices)} \\
&= (\mathbf{C}^\top)^\top \mathbf{T}^\top \mathbf{C}^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{C}\,\mathbf{T}^\top \mathbf{C}^\top && \text{(Definition 3 in Matrices)} \\
&= \mathbf{C}\mathbf{T}\mathbf{C}^\top, && (\mathbf{T} \text{ is symmetric}) \\
(\mathbf{GA})^\top &= (\mathbf{R}^\top \mathbf{S}\mathbf{R})^\top && \text{(second product)} \\
&= (\mathbf{R}^\top\,(\mathbf{S}\mathbf{R}))^\top && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{S}\mathbf{R})^\top\,(\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{S}^\top\,(\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{S}^\top\,\mathbf{R} && \text{(Definition 3 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}\mathbf{R}. && (\mathbf{S} \text{ is symmetric})
\end{aligned}
$$

Condition 1.

$$
\begin{aligned}
\mathbf{AGA} &= \mathbf{C}\mathbf{T}\mathbf{C}^\top \mathbf{C}\mathbf{R} && \text{(first product)} \\
&= \mathbf{C}\,\mathbf{T}\,(\mathbf{C}^\top \mathbf{C})\,\mathbf{R} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{C}\,\mathbf{I}_r\,\mathbf{R} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{A}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

Condition 2.

$$
\begin{aligned}
\mathbf{GAG} &= \mathbf{R}^\top \mathbf{S}\mathbf{R}\,\mathbf{R}^\top \mathbf{S}\mathbf{T}\mathbf{C}^\top && \text{(second product)} \\
&= \mathbf{R}^\top\,\mathbf{S}\,(\mathbf{R}\mathbf{R}^\top)\,\mathbf{S}\,\mathbf{T}\mathbf{C}^\top && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{I}_r\,\mathbf{S}\,\mathbf{T}\mathbf{C}^\top && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{G}. && \text{(Theorem 7 in Matrices)}
\end{aligned}
$$

□

Example 0.17 (The Moore-Penrose inverse of a rank-one matrix)**.** For $\mathbf{A}$ in Example 1 in Column Space, Null Space and Rank-Nullity, the rank factorization $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$ (Example 19 in Column Space, Null Space and Rank-Nullity) gives

$$\begin{aligned} \mathbf{C}^\top \mathbf{C} &= 1 + 9 \\ &= 10 \end{aligned}$$

and

$$\begin{aligned} \mathbf{R}\mathbf{R}^\top &= 1 + 4 + 4 \\ &= 9, \end{aligned}$$

so

$$\begin{aligned} \mathbf{A}^+ &= \frac{1}{9 \cdot 10} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \end{bmatrix} \\ &= \frac{1}{90} \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}. \end{aligned}$$

As a check on condition 1,

$$\begin{aligned} \mathbf{A}\mathbf{A}^+ &= \mathbf{C}\mathbf{T}\mathbf{C}^\top \\ &= \frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}, \end{aligned}$$

and its first row times $\mathbf{A}$ is $\frac{1}{10}(1 \cdot (1, -2, -2) + 3 \cdot (3, -6, -6)) = (1, -2, -2)$, the first row of $\mathbf{A}$; the second row is 3 times the first, like the second row of $\mathbf{A}$.

Definition 0.6 (Consistent linear system)**.** A linear system (Definition 12 in Matrices) $\mathbf{A}\tilde{x} = \tilde{b}$, with $\mathbf{A}$ an $m \times n$ matrix and $\tilde{b} \in \mathbb{R}^m$, is **consistent** if it has at least one solution $\tilde{x} \in \mathbb{R}^n$, and **inconsistent** otherwise.

Example 0.18 (A consistent system and an inconsistent one)**.** With $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\mathbf{A}(x_1, x_2) = (x_1, 0)$ (Definition 11 in Matrices).

- $\mathbf{A}\tilde{x} = (2, 0)$ is consistent: $\tilde{x} = (2, 0)$ is a solution.
- $\mathbf{A}\tilde{x} = (2, 1)$ is inconsistent: the second entry of $\mathbf{A}\tilde{x}$ is always 0, never 1.

Theorem 0.13 (When a linear system has a solution)**.** *Let* $\mathbf{A}$ *be* $m \times n$*, let* $\tilde{b} \in \mathbb{R}^m$*, and let* $\mathbf{G}$ *be any generalized inverse of* $\mathbf{A}$ *(Definition 0.4). The following statements are equivalent:*

1. $\mathbf{A}\tilde{x} = \tilde{b}$ *is consistent (Definition 0.6);*
2. $\tilde{b} \in \mathcal{C}(\mathbf{A})$ *(Definition 1 in Column Space, Null Space and Rank-Nullity);*
3. $\mathbf{A}\mathbf{G}\tilde{b} = \tilde{b}$*.*

Proof

Proof. **1 and 2 are equivalent.** Statement 1 says $\tilde{b} = \mathbf{A}\tilde{x}$ for some $\tilde{x}$ (Definition 0.6), and $\mathcal{C}(\mathbf{A})$ is the set of vectors $\mathbf{A}\tilde{x}$ (Definition 1 in Column Space, Null Space and Rank-Nullity), so $\tilde{b}$ is in it exactly when $\tilde{b} = \mathbf{A}\tilde{x}$ for some $\tilde{x}$.
1 implies 3. By Theorem 0.9, $\mathbf{G}\tilde{b}$ is a solution, so $\mathbf{A}\left(\mathbf{G}\tilde{b}\right) = \tilde{b}$, which is statement 3 (Theorem 5 in Matrices).
3 implies 1. If $\mathbf{A}\mathbf{G}\tilde{b} = \tilde{b}$, then $\tilde{x} = \mathbf{G}\tilde{b}$ is a solution, since $\mathbf{A}\left(\mathbf{G}\tilde{b}\right) = \left(\mathbf{A}\mathbf{G}\right)\tilde{b}$ (Theorem 5 in Matrices). □

Example 0.19 (Testing consistency with a generalized inverse)**.** For $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and the generalized inverse $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$, $\mathbf{A}\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$.

- $\tilde{b} = (2, 0)$:

$$\begin{aligned} \mathbf{A}\mathbf{G}\tilde{b} &= (2, 0) \\ &= \tilde{b}, \end{aligned}$$

  so the system is consistent.

- $\tilde{b} = (2, 1)$: $\mathbf{A}\mathbf{G}\tilde{b} = (7, 0) \neq \tilde{b}$, so it is not, as Example 0.13 found directly.

Theorem 0.14 (All solutions of a linear system)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *with a generalized inverse* $\mathbf{G}$ *(Definition 0.4).*

1. $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I}_n - \mathbf{G}\mathbf{A})$*:* $\tilde{x}$ *solves* $\mathbf{A}\tilde{x} = \tilde{0}_m$ *exactly when* $\tilde{x} = \left(\mathbf{I}_n - \mathbf{G}\mathbf{A}\right)\tilde{q}$ *for some* $\tilde{q} \in \mathbb{R}^n$*.*
2. *If* $\mathbf{A}\tilde{x} = \tilde{b}$ *is consistent (Definition 0.6), then* $\tilde{x}$ *is a solution exactly when*

$$\tilde{x} = \mathbf{G}\tilde{b} + \left(\mathbf{I}_n - \mathbf{G}\mathbf{A}\right)\tilde{q} \quad \text{for some } \tilde{q} \in \mathbb{R}^n :$$

one particular solution plus a vector of $\mathcal{N}(\mathbf{A})$.

Proof

Proof. **Part 1.** First,

$$\begin{aligned}
\mathbf{A}\left(\mathbf{I}_n - \mathbf{GA}\right) &= \mathbf{AI}_n - \mathbf{A}\left(\mathbf{GA}\right) && \text{(Theorem 0.7)}\\
&= \mathbf{AI}_n - \mathbf{AGA} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A} - \mathbf{AGA} && \text{(Theorem 7 in Matrices)}\\
&= \mathbf{A} - \mathbf{A} && \text{(Definition 0.4)}\\
&= \mathbf{0}_{m\times n}, && \text{(arithmetic)}
\end{aligned}$$

so $\mathbf{A}\left(\mathbf{I}_n - \mathbf{GA}\right)\tilde{q} = \tilde{0}_m$ for every $\tilde{q}$ (Theorem 5 in Matrices), and $\mathcal{C}(\mathbf{I}_n - \mathbf{GA}) \subseteq \mathcal{N}(\mathbf{A})$. Conversely, if $\mathbf{A}\tilde{x} = \tilde{0}_m$, then

$$\begin{aligned}
\left(\mathbf{I}_n - \mathbf{GA}\right)\tilde{x} &= \mathbf{I}_n\tilde{x} - \left(\mathbf{GA}\right)\tilde{x} && \text{(Theorem 0.7, with } \tilde{x} \text{ as the last factor)}\\
&= \tilde{x} - \left(\mathbf{GA}\right)\tilde{x} && \text{(Theorem 7 in Matrices)}\\
&= \tilde{x} - \mathbf{G}\left(\mathbf{A}\tilde{x}\right) && \text{(Theorem 5 in Matrices)}\\
&= \tilde{x} - \mathbf{G}\,\tilde{0}_m && (\mathbf{A}\tilde{x} = \tilde{0}_m)\\
&= \tilde{x}, && (\mathbf{G}\,\tilde{0}_m = \tilde{0}_n)
\end{aligned}$$

so $\tilde{x}$ is in $\mathcal{C}(\mathbf{I}_n - \mathbf{GA})$, with $\tilde{q} = \tilde{x}$.
Part 2. $\mathbf{G}\tilde{b}$ is a solution (Theorem 0.9). For any $\tilde{x}$,

$$\begin{aligned}
\mathbf{A}\left(\tilde{x} - \mathbf{G}\tilde{b}\right) &= \mathbf{A}\tilde{x} - \mathbf{A}\left(\mathbf{G}\tilde{b}\right) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1)\\
&= \mathbf{A}\tilde{x} - \mathbf{AG}\tilde{b} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{A}\tilde{x} - \tilde{b}, && \text{(Theorem 0.13, statement 3)}
\end{aligned}$$

so $\tilde{x}$ solves $\mathbf{A}\tilde{x} = \tilde{b}$ exactly when $\tilde{x} - \mathbf{G}\tilde{b} \in \mathcal{N}(\mathbf{A})$, that is, by part 1, exactly when $\tilde{x} - \mathbf{G}\tilde{b} = \left(\mathbf{I}_n - \mathbf{GA}\right)\tilde{q}$ for some $\tilde{q}$. □

Example 0.20 (All solutions of a consistent system)**.** For $\mathbf{A} = \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix}$, $\mathbf{G} = \begin{bmatrix}1 & 5\\ 7 & 0\end{bmatrix}$ and $\tilde{b} = (2, 0)$: $\mathbf{G}\tilde{b} = (2, 14)$, $\mathbf{GA} = \begin{bmatrix}1 & 0\\ 7 & 0\end{bmatrix}$, and $\mathbf{I}_2 - \mathbf{GA} = \begin{bmatrix}0 & 0\\ -7 & 1\end{bmatrix}$, so

$$\tilde{x} = (2, 14) + (0, -7q_1 + q_2), \qquad q_1, q_2 \in \mathbb{R}.$$

Since $-7q_1 + q_2$ can be any number, the solutions are exactly the vectors $(2, t)$, $t \in \mathbb{R}$,

which is what

$$\begin{aligned}\mathbf{A}\left(x_1, x_2\right) &= \left(x_1, 0\right) \\ &= \left(2, 0\right)\end{aligned}$$

says directly.

Corollary 0.1 (Solvable for every right-hand side, and solvable uniquely)**.** *Let* $\mathbf{A}$ *be* $m \times n$*.*

1. $\mathbf{A}\tilde{x} = \tilde{b}$ *is consistent for every* $\tilde{b} \in \mathbb{R}^m$ *exactly when* $\text{rank}(\mathbf{A}) = m$*.*
2. *A consistent system* $\mathbf{A}\tilde{x} = \tilde{b}$ *has exactly one solution exactly when* $\text{rank}(\mathbf{A}) = n$*.*

Proof

Proof. **Part 1.** By Theorem 0.13, the system is consistent for every $\tilde{b}$ exactly when $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$. $\mathcal{C}(\mathbf{A})$ is a subspace of $\mathbb{R}^m$ (Theorem 1 in Column Space, Null Space and Rank-Nullity) with dimension $\text{rank}(\mathbf{A})$ (Theorem 5 in Column Space, Null Space and Rank-Nullity), and $\dim(\mathbb{R}^m) = m$ (Example 16 in Subspaces and Rank). If the dimension is m, then $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ (Theorem 11 in Subspaces and Rank); if $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$, the dimension is m.
Part 2. By Theorem 0.14, the solutions are a particular solution plus the vectors of $\mathcal{N}(\mathbf{A})$, so there is exactly one solution exactly when $\mathcal{N}(\mathbf{A}) = \left\{\tilde{0}_n\right\}$, that is, when $\text{nullity}(\mathbf{A}) = 0$ (a subspace of dimension 0 has the empty list as a basis, whose span is $\left\{\tilde{0}\right\}$, by Definition 8 in Subspaces and Rank and Definition 5 in Subspaces and Rank; and $\left\{\tilde{0}\right\}$ has dimension 0, by Example 16 in Subspaces and Rank). By Theorem 6 in Column Space, Null Space and Rank-Nullity, that is when $\text{rank}(\mathbf{A}) = n$. □

Example 0.21 (Three systems)**.**

- The 2×3 matrix $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ of Example 13 in Column Space, Null Space and Rank-Nullity has rank $2 = m$, so every system with it is consistent, but $2 < 3 = n$, so the solutions are never unique.
- The 3×2 matrix $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ of Example 0.1 has rank $2 = n$, so a consistent system has one solution, but $2 < 3 = m$, so some systems are inconsistent: $(1, 0, 0) \notin \mathcal{C}(\mathbf{X})$, because $a\left(1, 1, 1\right) + b\left(1, 2, 3\right)$ has equal differences between consecutive entries, and $(1, 0, 0)$ does not.

- An invertible $p \times p$ matrix $\mathbf{A}$ has rank p: if $\mathbf{A}\tilde{x} = \tilde{0}$ then

$$\begin{aligned}\tilde{x} &= \mathbf{A}^{-1}\mathbf{A}\tilde{x} \\ &= \tilde{0},\end{aligned}$$

so its nullity is 0 and its rank is p (Theorem 6 in Column Space, Null Space and Rank-Nullity). So every system with it is consistent (part 1, with $p = m$) and has exactly one solution (part 2, with $p = n$), namely $\mathbf{A}^{-1}\tilde{b}$, since $\mathbf{A}\left(\mathbf{A}^{-1}\tilde{b}\right) = \tilde{b}$.

Theorem 0.15 (The projection onto a column space, with any generalized inverse)**.** *Let* $\mathbf{X}$ *be an* $n \times p$ *matrix of any rank, and let* $\mathbf{G}$ *be any generalized inverse of* $\mathbf{X}^\top\mathbf{X}$ *(Definition 0.4). Then for every* $\tilde{y} \in \mathbb{R}^n$, $\mathbf{X}\mathbf{G}\mathbf{X}^\top\tilde{y}$ *is the orthogonal projection of* $\tilde{y}$ *onto* $\mathcal{C}(\mathbf{X})$ *(Definition 0.3). So the matrix* $\mathbf{X}\mathbf{G}\mathbf{X}^\top$ *is the same for every choice of* $\mathbf{G}$*, and when* $\operatorname{rank}(\mathbf{X}) = p$ *it is the hat matrix* $\mathbf{H}$ *(Definition 0.2).*

Proof

Proof. **$\mathbf{X}^\top\tilde{y}$ is in $\mathcal{C}(\mathbf{X}^\top\mathbf{X})$.** $\mathcal{C}(\mathbf{X}^\top\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$, since $\left(\mathbf{X}^\top\mathbf{X}\right)\tilde{z} = \mathbf{X}^\top\left(\mathbf{X}\tilde{z}\right)$ (Theorem 5 in Matrices). Both are subspaces of $\mathbb{R}^p$ (Theorem 1 in Column Space, Null Space and Rank-Nullity) with the same dimension: $\operatorname{rank}(\mathbf{X}^\top\mathbf{X}) = \operatorname{rank}(\mathbf{X}^\top)$ (Theorem 8 in Column Space, Null Space and Rank-Nullity, Theorem 5 in Column Space, Null Space and Rank-Nullity). So $\mathcal{C}(\mathbf{X}^\top\mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ (Theorem 11 in Subspaces and Rank), and since $\mathbf{X}^\top\tilde{y} \in \mathcal{C}(\mathbf{X}^\top)$ (Definition 1 in Column Space, Null Space and Rank-Nullity), $\mathbf{X}^\top\tilde{y} = \mathbf{X}^\top\mathbf{X}\tilde{z}$ for some $\tilde{z} \in \mathbb{R}^p$.

The split. Let

$$\begin{aligned}\tilde{u} &= \mathbf{X}\mathbf{G}\mathbf{X}^\top\tilde{y} \\ &= \mathbf{X}\left(\mathbf{G}\mathbf{X}^\top\tilde{y}\right)\end{aligned}$$

(Theorem 5 in Matrices), which is in $\mathcal{C}(\mathbf{X})$ (Definition 1 in Column Space, Null Space and Rank-Nullity). For $\tilde{y} - \tilde{u}$,

$$
\begin{aligned}
\mathbf{X}^\top(\tilde{y}-\tilde{u}) &= \mathbf{X}^\top\tilde{y}-\mathbf{X}^\top\tilde{u} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1\text{)}\\
&= \mathbf{X}^\top\tilde{y}-\mathbf{X}^\top(\mathbf{X}\mathbf{G}\mathbf{X}^\top\tilde{y}) && \text{(substitute } \tilde{u}\text{)}\\
&= \mathbf{X}^\top\tilde{y}-(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\tilde{y}) && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{X}^\top\tilde{y}-(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X})\,\tilde{z} && \text{(first step)}\\
&= \mathbf{X}^\top\tilde{y}-((\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X}))\,\tilde{z} && \text{(Theorem 5 in Matrices)}\\
&= \mathbf{X}^\top\tilde{y}-(\mathbf{X}^\top\mathbf{X})\,\tilde{z} && \text{(Definition 0.4)}\\
&= \mathbf{X}^\top\tilde{y}-\mathbf{X}^\top\tilde{y} && \text{(first step)}\\
&= \tilde{0}_p, && \text{(arithmetic)}
\end{aligned}
$$

so $\tilde{y}-\tilde{u}\in\mathcal{N}(\mathbf{X}^\top)=\mathcal{C}(\mathbf{X})^\perp$ (Theorem 7 in Direct Sums and Orthogonal Complements). By the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements, $\tilde{u}$ is the orthogonal projection of $\tilde{y}$ onto $\mathcal{C}(\mathbf{X})$.

Same matrix. Every choice of $\mathbf{G}$ gives a matrix that projects every $\tilde{y}$ onto $\mathcal{C}(\mathbf{X})$, so all choices give the same matrix (Theorem 0.5, part 2). When $\text{rank}(\mathbf{X})=p$, $(\mathbf{X}^\top\mathbf{X})^{-1}$ is one choice (Theorem 0.1, Theorem 0.8), and it gives $\mathbf{H}$. □

Example 0.22 (A design matrix with a repeated column)**.** Let $\mathbf{X}=\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, an intercept column entered twice, so $\text{rank}(\mathbf{X})=1<2$ and $\mathbf{X}^\top\mathbf{X}=\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$ is not invertible: its rank is $\text{rank}(\mathbf{X})=1<2$ (Theorem 8 in Column Space, Null Space and Rank-Nullity), while an invertible 2×2 matrix has rank 2 (Example 0.21). The matrix $\mathbf{G}=\begin{bmatrix}\frac{1}{2} & 0\\ 0 & 0\end{bmatrix}$ is a generalized inverse of it:

$$
\begin{aligned}
(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}\,(\mathbf{X}^\top\mathbf{X}) &= \begin{bmatrix}1 & 0\\ 1 & 0\end{bmatrix}\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix} && \text{(Definition 7 in Matrices, for the first product)}\\
&= \begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}. && \text{(Definition 7 in Matrices)}
\end{aligned}
$$

Then

$$
\begin{aligned}
\mathbf{X}\mathbf{G}\mathbf{X}^\top &= \begin{bmatrix}\frac{1}{2} & 0\\ \frac{1}{2} & 0\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} && \text{(Definition 7 in Matrices, for } \mathbf{X}\mathbf{G}\text{)}\\
&= \begin{bmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix}, && \text{(Definition 7 in Matrices)}
\end{aligned}
$$

the hat matrix of the intercept-only model (Example 0.2). Another generalized inverse,

$\mathbf{G}' = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, which is one because $(\mathbf{X}^\top\mathbf{X})\,\mathbf{G}' = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$ and that times $\mathbf{X}^\top\mathbf{X}$ is again $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, gives $\mathbf{X}\mathbf{G}' = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$ and the same $\mathbf{X}\mathbf{G}'\mathbf{X}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, as Theorem 0.15 says it must. Both project onto $\mathcal{C}(\mathbf{X}) = \operatorname{span}\{(1,1)\}$, the column space of the intercept-only design.

Solving linear systems and least squares

i Source

This section is adapted from Zhou (2024b) and Zhou (2024a), used under the MIT License (see the license text in Section 2 in Subspaces and Rank). The first source states that every invertible matrix has an LU factorization; that is false without reordering rows (the 2×2 row swap $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has none), so this section states only how an LU factorization, when one exists, solves a system. The least squares results are proved here from the closest-point theorem (Theorem 0.3) rather than by differentiating, and the QR factorization is derived from Gram-Schmidt (Theorem 9 in Inner Products and Orthogonality).

Definition 0.7 (Triangular matrix)**.** A square matrix $\mathbf{U}$ is **upper triangular** if every entry below the diagonal is 0: $u_{ij} = 0$ whenever $i > j$. A square matrix $\mathbf{L}$ is **lower triangular** if every entry above the diagonal is 0: $\ell_{ij} = 0$ whenever $i < j$. A lower triangular matrix whose diagonal entries are all 1 is **unit lower triangular**.

Example 0.23 (Triangular and not)**.**

- $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ is upper triangular: its entries below the diagonal, in positions $(2,1)$, $(3,1)$ and $(3,2)$, are 0.
- $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$ is unit lower triangular.
- A diagonal matrix (Definition 4 in Special Matrices and Decompositions) is both upper and lower triangular.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is neither: its $(2,1)$ entry is not 0, and neither is its $(1,2)$ entry.

Theorem 0.16 (Solving a triangular system by substitution)**.** *Let* $\mathbf{U}$ *be an* $n \times n$ *upper triangular matrix (Definition 0.7) whose diagonal entries* $u_{11}, \dots, u_{nn}$ *are all nonzero. For every* $\tilde{b} \in \mathbb{R}^n$*, the system* $\mathbf{U}\tilde{x} = \tilde{b}$ *has exactly one solution, given for* $i = n, n-1, \dots, 1$ *in turn by*

$$x_i = \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j \right)$$

*(**back substitution**). Likewise, if* $\mathbf{L}$ *is lower triangular with nonzero diagonal,* $\mathbf{L}\tilde{y} = \tilde{b}$ *has exactly one solution, given for* $i = 1, \dots, n$ *by* $y_i = \frac{1}{\ell_{ii}} \left(b_i - \sum_{j=1}^{i-1} \ell_{ij}\, y_j \right)$ *(**forward substitution**).*

Proof

Proof. Row i of $\mathbf{U}\tilde{x} = \tilde{b}$ reads

$$\begin{aligned} b_i &= \sum_{j=1}^{n} u_{ij}\, x_j && \text{(Definition 11 in Matrices)} \\ &= \sum_{j=i}^{n} u_{ij}\, x_j && (u_{ij} = 0 \text{ for } j < i) \\ &= u_{ii}\, x_i + \sum_{j=i+1}^{n} u_{ij}\, x_j. && \text{(split off the } j = i \text{ term)} \end{aligned}$$

This equation is equivalent to each of

$$\begin{aligned} u_{ii}\, x_i &= b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j && \text{(subtract the sum from both sides)} \\ x_i &= \frac{1}{u_{ii}} \left(b_i - \sum_{j=i+1}^{n} u_{ij}\, x_j \right). && \text{(divide by } u_{ii} \neq 0) \end{aligned}$$

Row n involves only x_n, so it fixes x_n; once $x_{i+1}, \dots, x_n$ are fixed, row i fixes x_i. So the rows, taken from the last to the first, hold exactly when $\tilde{x}$ is the vector the formula builds: there is one solution, and only one. For $\mathbf{L}$, row i reads $\sum_{j=1}^{i-1} \ell_{ij}\, y_j + \ell_{ii}\, y_i = b_i$, and the same argument runs from the first row to the last. □

Example 0.24 (Back substitution on a 3×3 system)**.** Solve $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \tilde{x} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$:

1.
$$\begin{aligned} x_3 &= \frac{1}{-1} \cdot 1 \\ &= -1; \end{aligned}$$

2.
$$\begin{aligned} x_2 &= \frac{1}{1/2}\left(1 - \tfrac{1}{2} \cdot (-1)\right) \\ &= 2 \cdot \tfrac{3}{2} \\ &= 3; \end{aligned}$$

3.
$$\begin{aligned} x_1 &= \frac{1}{2}(8 - 1 \cdot 3 - (-1)(-1)) \\ &= \frac{1}{2} \cdot 4 \\ &= 2. \end{aligned}$$

So $\tilde{x} = (2, 3, -1)$. Without a nonzero diagonal the method fails: $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \tilde{x} = (1, 1)$ has no solution, since its second row reads $0 = 1$.

Theorem 0.17 (QR factorization)**.** *Let* $\mathbf{A}$ *be an* $m \times n$ *matrix with* $\operatorname{rank}(\mathbf{A}) = n$*. Then*

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times n}{\mathbf{Q}} \; \underset{n\times n}{\mathbf{R}},$$

where the columns of $\mathbf{Q}$ *are orthonormal (Definition 17 in Vectors) and* $\mathbf{R}$ *is upper triangular (Definition 0.7) with positive diagonal entries.*

Proof

Proof. The n columns $\tilde{a}_1, \ldots, \tilde{a}_n$ of $\mathbf{A}$ are linearly independent (Definition 3 in Subspaces and Rank), so the Gram-Schmidt process (Definition 10 in Inner Products and Orthogonality) runs all n steps and gives orthonormal $\tilde{q}_1, \ldots, \tilde{q}_n$ (Theorem 9 in Inner Products and Orthogonality, parts 1 and 3); let $\mathbf{Q}$ have these columns. At step i, $\tilde{\tilde{q}}_i \neq \tilde{0}$, and rearranging the orthogonalize step,

$$\begin{aligned}\tilde{a}_i &= \tilde{\tilde{q}}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i)\, \tilde{q}_j \qquad \text{(add the sum to both sides of the orthogonalize step)} \\ &= \left\|\tilde{\tilde{q}}_i\right\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i)\, \tilde{q}_j. \quad \text{(normalize step: } \tilde{\tilde{q}}_i = \left\|\tilde{\tilde{q}}_i\right\| \tilde{q}_i)\end{aligned}$$

Let $\mathbf{R}$ be the $n \times n$ matrix with $r_{ji} = \tilde{q}_j \cdot \tilde{a}_i$ for $j < i$, $r_{ii} = \left\|\tilde{\tilde{q}}_i\right\| > 0$, and $r_{ji} = 0$ for $j > i$; it is upper triangular with positive diagonal. Column i of $\mathbf{QR}$ is $\mathbf{Q}$ times column i of $\mathbf{R}$ (Definition 7 in Matrices), and

$$\begin{aligned}\mathbf{Q}\,(r_{1i}, \dots, r_{ni}) &= \sum_{j=1}^{n} r_{ji}\, \tilde{q}_j && \text{(Theorem 3 in Subspaces and Rank)} \\ &= \sum_{j=1}^{i} r_{ji}\, \tilde{q}_j && (r_{ji} = 0 \text{ for } j > i) \\ &= r_{ii}\, \tilde{q}_i + \sum_{j=1}^{i-1} r_{ji}\, \tilde{q}_j && \text{(split off the } j = i \text{ term)} \\ &= \left\|\tilde{\tilde{q}}_i\right\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i)\, \tilde{q}_j && \text{(the entries of } \mathbf{R}) \\ &= \tilde{a}_i. && \text{(the display)}\end{aligned}$$

So $\mathbf{QR} = \mathbf{A}$. □

Example 0.25 (A QR factorization from Gram-Schmidt)**.** For $\mathbf{A}$ with columns $\tilde{a}_1 = (1,1,0)$, $\tilde{a}_2 = (1,0,1)$, $\tilde{a}_3 = (0,1,1)$, Example 21 in Inner Products and Orthogonality and Example 23 in Inner Products and Orthogonality found $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1,1,0)$, $\tilde{q}_2 = \frac{1}{\sqrt{6}}(1,-1,2)$, $\tilde{q}_3 = \frac{1}{\sqrt{3}}(-1,1,1)$, with $\left\|\tilde{\tilde{q}}_1\right\| = \sqrt{2}$, $\left\|\tilde{\tilde{q}}_2\right\| = \sqrt{3/2}$, $\left\|\tilde{\tilde{q}}_3\right\| = 2/\sqrt{3}$,

$$\begin{aligned}\tilde{q}_1 \cdot \tilde{a}_2 &= \tilde{q}_1 \cdot \tilde{a}_3 \\ &= \tfrac{1}{\sqrt{2}}\end{aligned}$$

and $\tilde{q}_2 \cdot \tilde{a}_3 = \frac{1}{\sqrt{6}}$. So

$$\mathbf{R} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

As a check on the second column,

$$\begin{aligned}\tfrac{1}{\sqrt{2}}\,\tilde{q}_1 + \sqrt{3/2}\,\tilde{q}_2 &= \tfrac{1}{2}\,(1,1,0) + \tfrac{1}{2}\,(1,-1,2)\\ &= (1,0,1)\\ &= \tilde{a}_2.\end{aligned}$$

Definition 0.8 (Least squares solution)**.** Let $\mathbf{A}$ be $m \times n$ and $\tilde{b} \in \mathbb{R}^m$. A **least squares solution** of $\mathbf{A}\tilde{x} = \tilde{b}$ is a vector $\hat{\tilde{x}} \in \mathbb{R}^n$ such that

$$\left\|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\right\| \le \left\|\tilde{b} - \mathbf{A}\tilde{x}\right\| \quad \text{for every } \tilde{x} \in \mathbb{R}^n.$$

The equations $\mathbf{A}^\top\mathbf{A}\tilde{x} = \mathbf{A}^\top\tilde{b}$ are the **normal equations**.

Example 0.26 (Fitting a constant)**.** Let $\mathbf{A} = \begin{bmatrix}1\\1\end{bmatrix}$ and $\tilde{b} = (1,3)$. The system $x\,(1,1) = (1,3)$ is inconsistent (Definition 0.6): its first entry needs $x = 1$ and its second needs $x = 3$. For any number x,

$$\begin{aligned}\left\|\tilde{b} - \mathbf{A}x\right\|^2 &= (1-x)^2 + (3-x)^2\\ &= 2\,(x-2)^2 + 2,\end{aligned}$$

which is smallest at $x = 2$, so $\hat{x} = 2$ is the least squares solution. The value $x = 1$ is not one: it gives $0 + 4 = 4 > 2$. The normal equations read $2x = 4$, and their solution is $x = 2$ too.

Theorem 0.18 (Least squares solutions solve the normal equations)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *and* $\tilde{b} \in \mathbb{R}^m$*.*

1. $\hat{\tilde{x}}$ *is a least squares solution (Definition 0.8) exactly when it solves the normal equations* $\mathbf{A}^\top\mathbf{A}\hat{\tilde{x}} = \mathbf{A}^\top\tilde{b}$*.*
2. *The normal equations are consistent (Definition 0.6).*
3. *Every least squares solution gives the same fitted vector* $\mathbf{A}\hat{\tilde{x}}$*: the orthogonal projection of* $\tilde{b}$ *onto* $\mathcal{C}(\mathbf{A})$ *(Definition 0.3).*
4. *The least squares solution is unique exactly when* $\operatorname{rank}(\mathbf{A}) = n$*, and then* $\hat{\tilde{x}} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\tilde{b}$*.*

Proof

Proof. Let $\tilde{u}$ be the orthogonal projection of $\tilde{b}$ onto $\mathcal{C}(\mathbf{A})$. The vectors $\mathbf{A}\tilde{x}$, as $\tilde{x}$ ranges over $\mathbb{R}^n$, are exactly the points of $\mathcal{C}(\mathbf{A})$ (Definition 1 in Column Space, Null Space and

Rank-Nullity). **Least squares solutions are the solutions of $\mathbf{A}\tilde{x} = \tilde{u}$.** If $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$, then for every $\tilde{x}$, $\mathbf{A}\tilde{x} \in \mathcal{C}(\mathbf{A})$, so $\left\|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\right\| = \left\|\tilde{b} - \tilde{u}\right\| \leq \left\|\tilde{b} - \mathbf{A}\tilde{x}\right\|$ (Theorem 0.3), and $\hat{\tilde{x}}$ is a least squares solution. Conversely, let $\hat{\tilde{x}}$ be a least squares solution. $\tilde{u} \in \mathcal{C}(\mathbf{A})$, so $\tilde{u} = \mathbf{A}\tilde{x}_0$ for some $\tilde{x}_0$ (Definition 1 in Column Space, Null Space and Rank-Nullity), and

$$\begin{aligned}
\left\|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\right\| &\leq \left\|\tilde{b} - \mathbf{A}\tilde{x}_0\right\| && \text{(Definition 0.8)} \\
&= \left\|\tilde{b} - \tilde{u}\right\| && (\tilde{u} = \mathbf{A}\tilde{x}_0) \\
&\leq \left\|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\right\|, && \text{(Theorem 0.3, with } \tilde{w} = \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A}))
\end{aligned}$$

so the last inequality is an equality, and the equality case of Theorem 0.3 gives $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$.
Parts 1 and 3. $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ holds exactly when $\tilde{b} - \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})^{\perp}$: if $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$, then $\tilde{b} - \tilde{u} \in \mathcal{C}(\mathbf{A})^{\perp}$ (Definition 0.3); conversely, if $\tilde{b} - \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})^{\perp}$, then $\tilde{b} = \mathbf{A}\hat{\tilde{x}} + (\tilde{b} - \mathbf{A}\hat{\tilde{x}})$ is a split into $\mathcal{C}(\mathbf{A})$ and $\mathcal{C}(\mathbf{A})^{\perp}$, so $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ by the uniqueness in Theorem 8 in Direct Sums and Orthogonal Complements. And $\mathcal{C}(\mathbf{A})^{\perp} = \mathcal{N}(\mathbf{A}^{\top})$ (Theorem 7 in Direct Sums and Orthogonal Complements), where

$$\begin{aligned}
\mathbf{A}^{\top}\left(\tilde{b} - \mathbf{A}\hat{\tilde{x}}\right) &= \mathbf{A}^{\top}\tilde{b} - \mathbf{A}^{\top}\left(\mathbf{A}\hat{\tilde{x}}\right) && \text{(Theorem 0.7)} \\
&= \mathbf{A}^{\top}\tilde{b} - \mathbf{A}^{\top}\mathbf{A}\hat{\tilde{x}}, && \text{(Theorem 5 in Matrices)}
\end{aligned}$$

which is $\tilde{0}_n$ exactly when the normal equations hold.
Part 2. The vector $\tilde{x}_0$ above satisfies $\mathbf{A}\tilde{x}_0 = \tilde{u}$, so it is a least squares solution, and it solves the normal equations by part 1.
Part 4. The least squares solutions are exactly the solutions of $\mathbf{A}\tilde{x} = \tilde{u}$, a consistent system, so there is exactly one when $\operatorname{rank}(\mathbf{A}) = n$ and more than one otherwise (Corollary 0.1). When $\operatorname{rank}(\mathbf{A}) = n$, $\mathbf{A}^{\top}\mathbf{A}$ is invertible (Theorem 0.1), and with $\mathbf{M} \stackrel{\text{def}}{=} \mathbf{A}^{\top}\mathbf{A}$,

$$\begin{aligned}
\hat{\tilde{x}} &= \mathbf{I}_n\,\hat{\tilde{x}} && \text{(Theorem 7 in Matrices)} \\
&= \left(\mathbf{M}^{-1}\mathbf{M}\right)\hat{\tilde{x}} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{M}^{-1}\left(\mathbf{M}\hat{\tilde{x}}\right) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{M}^{-1}\,\mathbf{A}^{\top}\tilde{b}. && \text{(the normal equations)}
\end{aligned}$$

□

Example 0.27 (A least squares line)**.** Fit $y = x_1 + x_2 t$ to the points $(t, y) = (1,1), (2,2), (3,2)$: $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ (the $\mathbf{X}$ of Example 0.1) and $\tilde{b} = (1,2,2)$. $\mathbf{A}$ has rank 2 (Example 2 in Subspaces and Rank), so the solution is unique (part 4). $\mathbf{A}^{\top}\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$

has inverse $\frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$ (Example 0.1), and

$$\begin{aligned} \mathbf{A}^\top \tilde{b} &= (1+2+2,\ 1+4+6) \\ &= (5, 11), \end{aligned}$$

so

$$\begin{aligned} \hat{\tilde{x}} &= \frac{1}{6}(14 \cdot 5 - 6 \cdot 11,\ -6 \cdot 5 + 3 \cdot 11) \\ &= \frac{1}{6}(4, 3) \\ &= \left(\tfrac{2}{3}, \tfrac{1}{2}\right). \end{aligned}$$

The fitted values are $\mathbf{A}\hat{\tilde{x}} = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)$, and the residuals $\tilde{b} - \mathbf{A}\hat{\tilde{x}} = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)$ are orthogonal to both columns of $\mathbf{A}$: $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$ and $-\frac{1}{6} + \frac{2}{3} - \frac{1}{2} = 0$.

Example 0.28 (Many least squares solutions, one fitted vector)**.** With the rank-1 matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$ (Example 2 in Subspaces and Rank) and $\tilde{b} = (1, 2, 2)$, $\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$ and $\mathbf{A}^\top\tilde{b} = (5, 10)$, so both normal equations say $3x_1 + 6x_2 = 5$, that is, $x_1 + 2x_2 = \frac{5}{3}$. There are infinitely many least squares solutions, such as $(\frac{5}{3}, 0)$ and $(0, \frac{5}{6})$, as part 4 predicts for $\operatorname{rank}(\mathbf{A}) = 1 < 2$; but every one gives the same fitted vector

$$\begin{aligned} \mathbf{A}\hat{\tilde{x}} &= (x_1 + 2x_2)\,(1, 1, 1) \\ &= \tfrac{5}{3}\,(1, 1, 1), \end{aligned}$$

as part 3 says.

Theorem 0.19 (Solving least squares by QR)**.** *Let* $\mathbf{A}$ *be* $m \times n$ *with* $\operatorname{rank}(\mathbf{A}) = n$*, with QR factorization* $\mathbf{A} = \mathbf{QR}$ *(Theorem 0.17), and let* $\tilde{b} \in \mathbb{R}^m$*. The least squares solution (Definition 0.8) is the unique solution of*

$$\underbrace{\mathbf{R}}_{n\times n}\ \hat{\tilde{x}} = \underbrace{\mathbf{Q}^\top}_{n\times m}\ \tilde{b},$$

which back substitution computes (Theorem 0.16). When $m = n$*, it is the unique solution of* $\mathbf{A}\tilde{x} = \tilde{b}$*.*

Proof

Proof. $\mathbf{R}$ is upper triangular with positive diagonal, so $\mathbf{R}\tilde{x} = \mathbf{Q}^\top \tilde{b}$ has exactly one solution $\hat{\tilde{x}}$ (Theorem 0.16). $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_n$, as in the proof of Theorem 0.4. Then

$$\begin{aligned}
\mathbf{A}^\top\mathbf{A}\hat{\tilde{x}} &= (\mathbf{QR})^\top (\mathbf{QR})\,\hat{\tilde{x}} && \text{(substitute } \mathbf{A} = \mathbf{QR}\text{)} \\
&= \mathbf{R}^\top\,\mathbf{Q}^\top\,(\mathbf{QR})\,\hat{\tilde{x}} && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top\,(\mathbf{Q}^\top\mathbf{Q})\,(\mathbf{R}\hat{\tilde{x}}) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top\,\mathbf{I}_n\,(\mathbf{R}\hat{\tilde{x}}) && (\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_n) \\
&= \mathbf{R}^\top\,(\mathbf{R}\hat{\tilde{x}}) && \text{(Theorem 7 in Matrices)} \\
&= \mathbf{R}^\top\,(\mathbf{Q}^\top\tilde{b}) && (\mathbf{R}\hat{\tilde{x}} = \mathbf{Q}^\top\tilde{b}) \\
&= (\mathbf{R}^\top\,\mathbf{Q}^\top)\,\tilde{b} && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{QR})^\top\,\tilde{b} && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{A}^\top\tilde{b}, && (\mathbf{A} = \mathbf{QR})
\end{aligned}$$

so $\hat{\tilde{x}}$ solves the normal equations and is the least squares solution (Theorem 0.18), which is unique because $\operatorname{rank}(\mathbf{A}) = n$. When $m = n$,

$$\begin{aligned}
\operatorname{rank}(\mathbf{A}) &= n \\
&= m,
\end{aligned}$$

so $\mathbf{A}\tilde{x} = \tilde{b}$ is consistent for every $\tilde{b}$ (Corollary 0.1, part 1), so its solution leaves residual $\tilde{0}$, is a least squares solution, and is therefore $\hat{\tilde{x}}$. □

Example 0.29 (The least squares line by QR)**.** For $\mathbf{A}$ and $\tilde{b}$ of Example 0.27, Gram-Schmidt gives $\tilde{q}_1 = \frac{1}{\sqrt{3}}(1,1,1)$; then

$$\begin{aligned}
\tilde{q}_1 \cdot (1,2,3) &= \tfrac{6}{\sqrt{3}} \\
&= 2\sqrt{3},
\end{aligned}$$

$$\begin{aligned}
\tilde{\tilde{q}}_2 &= (1,2,3) - 2\,(1,1,1) \\
&= (-1,0,1)
\end{aligned}$$

with norm $\sqrt{2}$, and $\tilde{q}_2 = \frac{1}{\sqrt{2}}(-1,0,1)$. So

$$\mathbf{R} = \begin{bmatrix} \sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix},$$
$$\mathbf{Q}^{\top}\tilde{b} = \left(\tfrac{1+2+2}{\sqrt{3}},\ \tfrac{-1+0+2}{\sqrt{2}}\right) = \left(\tfrac{5}{\sqrt{3}}, \tfrac{1}{\sqrt{2}}\right).$$

Back substitution gives

$$x_2 = \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}} = \tfrac{1}{2}$$

and

$$x_1 = \tfrac{1}{\sqrt{3}}\left(\tfrac{5}{\sqrt{3}} - 2\sqrt{3} \cdot \tfrac{1}{2}\right) = \tfrac{5}{3} - 1 = \tfrac{2}{3},$$

the solution found in Example 0.27.

Definition 0.9 (LU factorization)**.** An **LU factorization** of an $n \times n$ matrix $\mathbf{A}$ is a product $\mathbf{A} = \mathbf{LU}$ with $\mathbf{L}$ unit lower triangular and $\mathbf{U}$ upper triangular (Definition 0.7).

Example 0.30 (An LU factorization of a 3×3 matrix)**.** Let $\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$, $\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}$ (unit lower triangular) and $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$ (upper triangular). By Definition 7 in Matrices, entry (i, j) of $\mathbf{LU}$ is $\sum_k \ell_{ik}\, u_{kj}$, so row i of $\mathbf{LU}$ is $\sum_k \ell_{ik}$ times row k of $\mathbf{U}$. Row 1 is $1 \cdot (2, 1, -1) = (2, 1, -1)$; row 2 is $-\frac{3}{2}(2, 1, -1) + (0, \frac{1}{2}, \frac{1}{2}) = (-3, -1, 2)$, and row 3 is $-(2, 1, -1) + 4(0, \frac{1}{2}, \frac{1}{2}) + (0, 0, -1) = (-2, 1, 2)$. These are the rows of $\mathbf{A}$, so $\mathbf{LU} = \mathbf{A}$ is an LU factorization. The below-diagonal entries of $\mathbf{L}$ are the negatives of the multiples of earlier rows that Gaussian elimination adds to reduce $\mathbf{A}$ to $\mathbf{U}$.

Example 0.31 (An invertible matrix with no LU factorization)**.** $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible: $\mathbf{P}^2 = \mathbf{I}_2$, so $\mathbf{P}^{-1} = \mathbf{P}$ (Definition 5 in Special Matrices and Decompositions); but it has no LU factorization. If $\mathbf{P} = \mathbf{LU}$, the $(1, 1)$ entry gives $0 = 1 \cdot u_{11}$, so $u_{11} = 0$, and then the $(2, 1)$ entry gives

$$\begin{aligned} 1 &= \ell_{21}\, u_{11} \\ &= 0, \end{aligned}$$

which is impossible. Swapping the two rows first removes the obstacle: the swapped matrix is $\mathbf{I}_2 = \mathbf{I}_2\,\mathbf{I}_2$.

Theorem 0.20 (Solving a system with an LU factorization)**.** *If $\mathbf{A} = \mathbf{LU}$ is an LU factorization (Definition 0.9) and the diagonal entries of $\mathbf{U}$ are all nonzero, then for every $\tilde{b} \in \mathbb{R}^n$ the system $\mathbf{A}\tilde{x} = \tilde{b}$ has exactly one solution: solve $\mathbf{L}\tilde{y} = \tilde{b}$ by forward substitution, then $\mathbf{U}\tilde{x} = \tilde{y}$ by back substitution (Theorem 0.16).*

Proof

Proof. $\mathbf{L}$ has diagonal entries 1 and $\mathbf{U}$ has nonzero ones, so each of the two triangular systems has exactly one solution (Theorem 0.16). If $\tilde{y}$ and $\tilde{x}$ are those solutions, then

$$\begin{aligned} \mathbf{A}\tilde{x} &= \mathbf{L}\,(\mathbf{U}\tilde{x}) \\ &= \mathbf{L}\tilde{y} \\ &= \tilde{b} \end{aligned}$$

(Theorem 5 in Matrices), so $\tilde{x}$ is a solution. Conversely, if $\mathbf{A}\tilde{x} = \tilde{b}$, then $\tilde{y} \overset{\text{def}}{=} \mathbf{U}\tilde{x}$ satisfies

$$\begin{aligned} \mathbf{L}\tilde{y} &= \mathbf{L}\,(\mathbf{U}\tilde{x}) \\ &= (\mathbf{LU})\,\tilde{x} \\ &= \mathbf{A}\tilde{x} \\ &= \tilde{b} \end{aligned}$$

(Theorem 5 in Matrices), so $\tilde{y}$ is the unique solution of that system, and $\tilde{x}$ is then the unique solution of $\mathbf{U}\tilde{x} = \tilde{y}$. □

Example 0.32 (Two triangular solves)**.** With $\mathbf{L}$ and $\mathbf{U}$ of Example 0.30, solve $\mathbf{A}\tilde{x} = (8, -11, -3)$. Forward substitution on $\mathbf{L}\tilde{y} = (8, -11, -3)$ gives $y_1 = 8$,

$$\begin{aligned} y_2 &= -11 + \tfrac{3}{2} \cdot 8 \\ &= 1, \end{aligned}$$

$$\begin{aligned} y_3 &= -3 + 8 - 4 \cdot 1 \\ &= 1. \end{aligned}$$

Back substitution on $\mathbf{U}\tilde{x} = (8, 1, 1)$ is Example 0.24, which gives $\tilde{x} = (2, 3, -1)$. As a check,

$$\begin{aligned}\mathbf{A}\,(2,3,-1) &= (4+3+1,\ -6-3-2,\ -4+3-2)\\ &= (8,-11,-3).\end{aligned}$$

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